Ex 4

Aim

To perform data inspection, filtering, statistical analysis, and correlation as part of Exploratory Data Analysis (EDA) using data science frameworks like CRISP-DM, SEMMA, and KDD.

Procedure

- **Data Inspection**: Checked dataset shape, column datatypes, missing values, and overall structure using .info(), .head(), and .shape.
- **Data Filtering**: Filtered out missing entries (dropna()), removed duplicate records (drop_duplicates()), and extracted subset conditions like transactions with Quantity > 10.
- **Statistical Analysis & Correlation**: Generated summary statistics (.describe()) to compute central tendencies, standard deviations, and quartiles, and calculated correlation matrices between numeric variables.

Program:

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("/content/OnlineRetail.csv", encoding='latin1')


print(df.head())

print(df.shape)

print(df.dtypes)

print(df.describe())
```

```
  InvoiceNo StockCode                          Description  Quantity  \
0    536365    85123A   WHITE HANGING HEART T-LIGHT HOLDER         6
1    536365     71053                  WHITE METAL LANTERN         6
2    536365    84406B       CREAM CUPID HEARTS COAT HANGER         8
3    536365    84029G  KNITTED UNION FLAG HOT WATER BOTTLE         6
4    536365    84029E       RED WOOLLY HOTTIE WHITE HEART.         6

      InvoiceDate  UnitPrice  CustomerID         Country
0  12/1/2010 8:26       2.55     17850.0  United Kingdom
1  12/1/2010 8:26       3.39     17850.0  United Kingdom
2  12/1/2010 8:26       2.75     17850.0  United Kingdom
3  12/1/2010 8:26       3.39     17850.0  United Kingdom
4  12/1/2010 8:26       3.39     17850.0  United Kingdom
(541909, 8)
InvoiceNo       object
StockCode       object
Description     object
Quantity         int64
InvoiceDate     object
UnitPrice      float64
CustomerID     float64
Country         object
```

```python
df["TotalSales"] = df["Quantity"] * df["UnitPrice"]
print(df.head())
```

Python

```
  InvoiceNo StockCode                          Description  Quantity  \
0    536365    85123A   WHITE HANGING HEART T-LIGHT HOLDER         6
1    536365     71053                  WHITE METAL LANTERN         6
2    536365    84406B       CREAM CUPID HEARTS COAT HANGER         8
3    536365    84029G  KNITTED UNION FLAG HOT WATER BOTTLE         6
4    536365    84029E       RED WOOLLY HOTTIE WHITE HEART.         6

      InvoiceDate  UnitPrice  CustomerID         Country  TotalSales
0  12/1/2010 8:26       2.55     17850.0  United Kingdom       15.30
1  12/1/2010 8:26       3.39     17850.0  United Kingdom       20.34
2  12/1/2010 8:26       2.75     17850.0  United Kingdom       22.00
3  12/1/2010 8:26       3.39     17850.0  United Kingdom       20.34
4  12/1/2010 8:26       3.39     17850.0  United Kingdom       20.34
```

```python
canceled = df[df["InvoiceNo"].astype(str).str.startswith("C", na=False)]
print("\nCanceled Transactions:")
print(canceled.head())
uk_orders = df[df["Country"] == "United Kingdom"]
print("\nOrders from United Kingdom:")
print(uk_orders.head())
```

Python

```
Canceled Transactions:
    InvoiceNo StockCode                        Description  Quantity  \
141   C536379         D                           Discount        -1
154   C536383    35004C    SET OF 3 COLOURED  FLYING DUCKS        -1
235   C536391     22556     PLASTERS IN TIN CIRCUS PARADE        -12
236   C536391     21984   PACK OF 12 PINK PAISLEY TISSUES        -24
237   C536391     21983   PACK OF 12 BLUE PAISLEY TISSUES        -24

         InvoiceDate  UnitPrice  CustomerID         Country  TotalSales
141   12/1/2010 9:41      27.50     14527.0  United Kingdom      -27.50
154   12/1/2010 9:49       4.65     15311.0  United Kingdom       -4.65
235  12/1/2010 10:24       1.65     17548.0  United Kingdom      -19.80
236  12/1/2010 10:24       0.29     17548.0  United Kingdom       -6.96
237  12/1/2010 10:24       0.29     17548.0  United Kingdom       -6.96

Orders from United Kingdom:
  InvoiceNo StockCode                          Description  Quantity  \
0    536365    85123A   WHITE HANGING HEART T-LIGHT HOLDER         6
1    536365     71053                  WHITE METAL LANTERN         6
2    536365    84406B       CREAM CUPID HEARTS COAT HANGER         8
3    536365    84029G  KNITTED UNION FLAG HOT WATER BOTTLE         6
4    536365    84029E       RED WOOLLY HOTTIE WHITE HEART.         6

      InvoiceDate  UnitPrice  CustomerID         Country  TotalSales
...
1  12/1/2010 8:26       3.39     17850.0  United Kingdom       20.34
2  12/1/2010 8:26       2.75     17850.0  United Kingdom       22.00
3  12/1/2010 8:26       3.39     17850.0  United Kingdom       20.34
4  12/1/2010 8:26       3.39     17850.0  United Kingdom       20.34
```

```python
# Aggregate & Bar Plot (Top 10 Countries by Total Sales)
country_sales = (
    df.groupby("Country")["TotalSales"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)
print("\nTop 10 Countries by Total Sales:")
print(country_sales)

plt.figure(figsize=(10, 5))
country_sales.plot(kind="bar", color="steelblue")
plt.title("Top 10 Countries by Total Sales")
plt.ylabel("Total Sales")
plt.xlabel("Country")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
Top 10 Countries by Total Sales:
Country
United Kingdom    8187806.364
Netherlands        284661.540
EIRE               263276.820
Germany            221698.210
France             197403.900
Australia          137077.270
Switzerland         56385.350
Spain               54774.580
Belgium             40910.960
Sweden              36595.910
Name: TotalSales, dtype: float64
```

```python
# Sort by highest TotalSales
sorted_df = df.sort_values(by="TotalSales", ascending=False)
print("\nSorted by Highest TotalSales:")
print(sorted_df.head())
```

```
Sorted by Highest TotalSales:
       InvoiceNo  StockCode                      Description  Quantity  \
540421    581483      23843      PAPER CRAFT , LITTLE BIRDIE     80995   
61619     541431      23166   MEDIUM CERAMIC TOP STORAGE JAR     74215   
222680    556444      22502  PICNIC BASKET WICKER 60 PIECES        60   
15017     537632  AMAZONFEE                       AMAZON FEE         1   
299982   A563185          B                  Adjust bad debt         1   

            InvoiceDate  UnitPrice  CustomerID         Country  TotalSales  
540421    12/9/2011 9:15       2.08     16446.0  United Kingdom   168469.60  
61619    1/18/2011 10:01       1.04     12346.0  United Kingdom    77183.60  
222680   6/10/2011 15:28     649.50     15098.0  United Kingdom    38970.00  
15017    12/7/2010 15:08   13541.33         NaN  United Kingdom    13541.33  
299982   8/12/2011 14:50   11062.06         NaN  United Kingdom    11062.06  
```

Central Tendency & Dispersion Metrics

```python
country_sales = (
    df.groupby("Country")["TotalSales"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)
print("\nTop 10 Countries by Total Sales:")
print(country_sales)

plt.figure(figsize=(10, 5))
country_sales.plot(kind="bar", color="steelblue")
plt.title("Top 10 Countries by Total Sales")
plt.ylabel("Total Sales")
plt.xlabel("Country")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
Top 10 Countries by Total Sales:
Country
United Kingdom    8187806.364
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Australia          137077.270
Switzerland         56385.350
Spain               54774.580
Belgium             40910.960
Sweden              36595.910
Name: TotalSales, dtype: float64
```
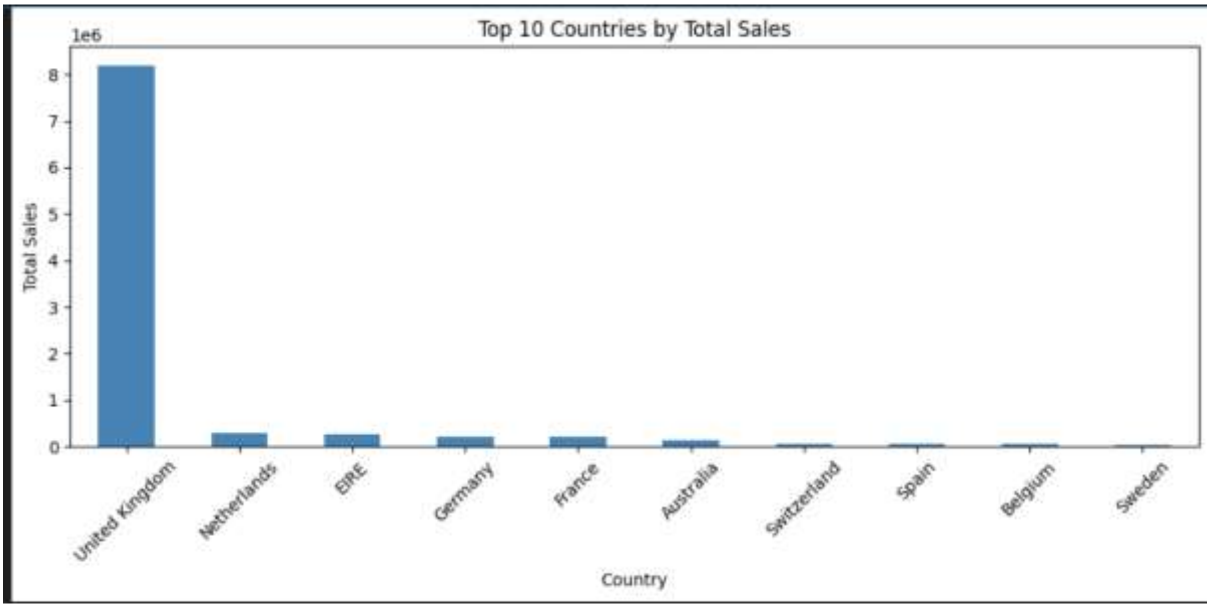
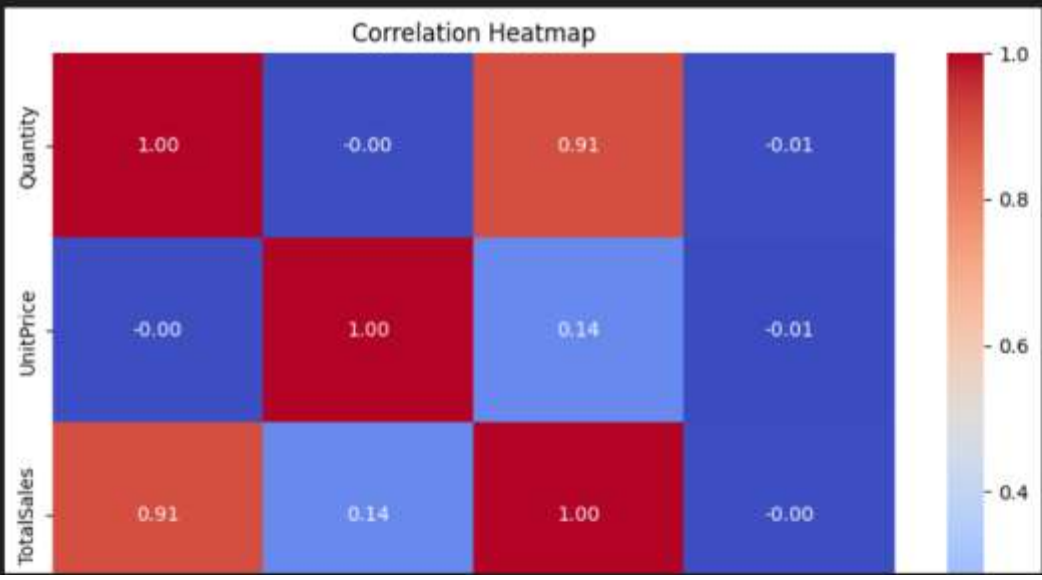



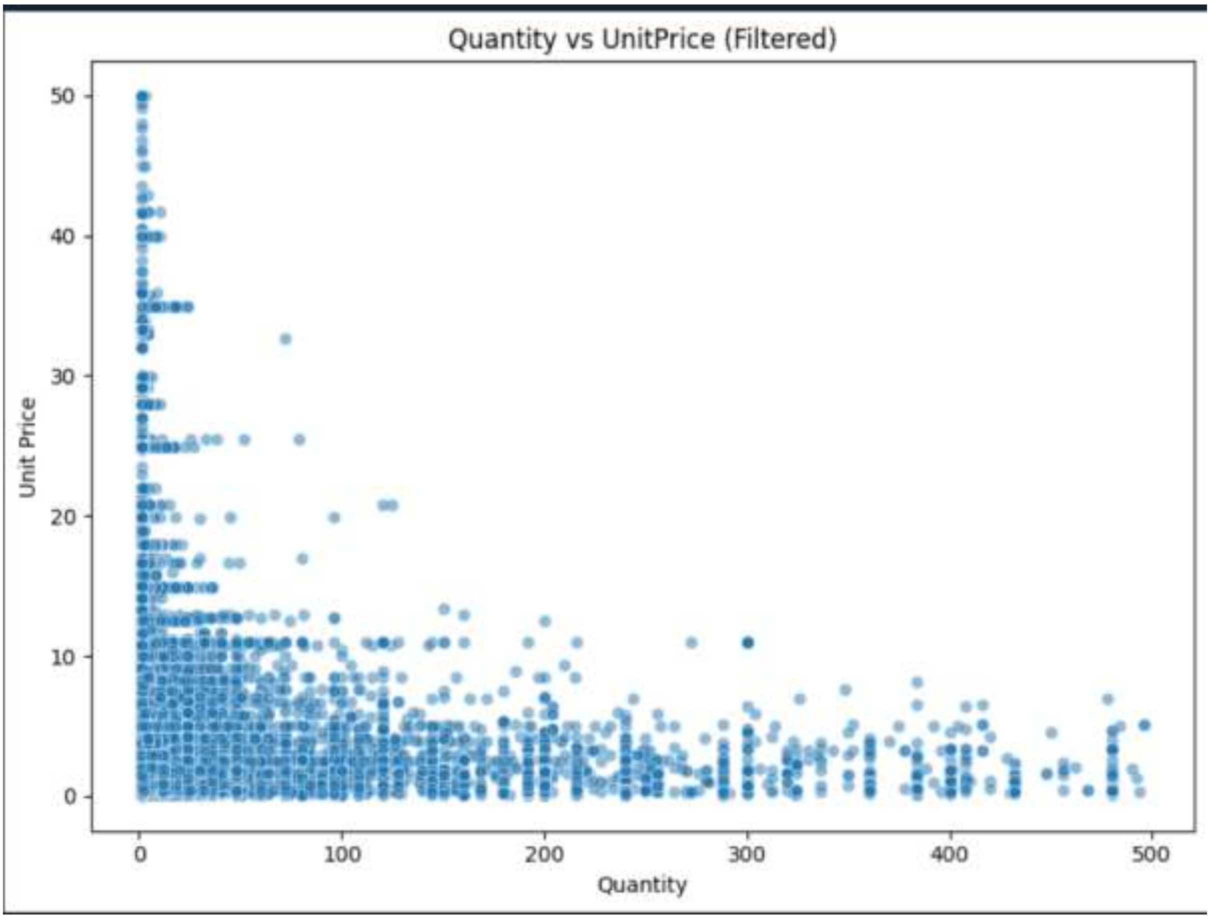
```python
valid_sales = df[(df["Quantity"] > 0) & (df["UnitPrice"] > 0)]
corr = valid_sales[["Quantity", "UnitPrice", "TotalSales", "CustomerID"]].corr()
print("\nCorrelation Matrix:")
print(corr)
plt.figure(figsize=(8, 6))
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()
```

[12]

```
Correlation Matrix:
            Quantity  UnitPrice  TotalSales  CustomerID
Quantity    1.000000  -0.003773    0.907338   -0.006232
UnitPrice  -0.003773   1.000000    0.137404   -0.010873
TotalSales  0.907338   0.137404    1.000000   -0.004109
CustomerID -0.006232  -0.010873   -0.004109    1.000000
```




```python
plt.figure(figsize=(8, 6))
sns.scatterplot(
    data=valid_sales[(valid_sales["Quantity"] < 500) & (valid_sales["UnitPrice"] < 50)],
    x="Quantity",
    y="UnitPrice",
    alpha=0.5
)
plt.title("Quantity vs UnitPrice (Filtered)")
plt.xlabel("Quantity")
plt.ylabel("Unit Price")
plt.tight_layout()
plt.show()
```



Result Thus the EDA, data inspection, filtering, statistical analysis, and correlation using data science frameworks like CRISP-DM, SEMMA, KDD is implemented successfully.